Entity Relationship Diagram

Main Entities

The **RISING WATER** flood prediction system is designed using four major entities: **User**, **Weather Data**, **Machine Learning Model**, and **Prediction**. These entities form the core of the database and work together to collect weather information, process it using machine learning algorithms, and generate accurate flood prediction results. Each entity contains specific attributes that describe its characteristics and is connected to other entities through well-defined relationships.

1. User Entity

The **User** entity stores the details of all registered users who access the flood prediction system. Users can log into the application, submit weather information, and view the flood prediction results. Each user is uniquely identified by the **User ID**, which acts as the primary key.

Attributes:

- User ID : U001
- User Name : Gyansi Pravallika
- Email :Gyansi@gmail.com
- Password : Gyansi123
- Role : Administrator

The **Role** attribute identifies the type of user, such as Administrator or General User, allowing the system to provide appropriate access permissions and security.

2. Weather Data Entity

The **Weather Data** entity stores the climatic and environmental information entered by users. These weather parameters are used as input by the machine learning model to predict flood probability.

Attributes:

- Weather Data ID : WD001
- User ID : U002
- Annual Rainfall : 980
- Cloud Visibility : 75
- Seasonal Rainfall : 320
- Temperature : 29
- Humidity : 78

The **User ID** serves as a foreign key that links each weather record to the user who submitted it. This entity contains all the necessary weather parameters required for flood prediction.

3. Machine Learning Model Entity

The **Machine Learning Model** entity stores information about the trained machine learning model used to generate flood predictions. It keeps track of the model details, algorithm used, prediction accuracy, and the trained model file.

Attributes:

- Model ID : M003
- Model Name : Flood Prediction Model
- Algorithm Type : Random Forest
- Accuracy : 95.6
- Model File : flood_prediction_model.pkl

This entity allows the system to manage different machine learning models and compare their performance. It also provides flexibility for updating or replacing the prediction model whenever improved algorithms become available.

4. Prediction Entity

The **Prediction** entity stores the results generated by the machine learning model after analyzing the submitted weather data. Each prediction record represents the flood prediction for a particular weather dataset.

Attributes:

- Prediction ID : P001
- Data ID : WD001
- Model ID : M001
- Prediction Result : Low flood risk
- Flood Probability : 18.5
- Prediction Date : 5-7-2025

The **Data ID** links the prediction to the corresponding weather data, while the **Model ID** identifies the machine learning model used to generate the prediction. This entity stores the final prediction outcome along with the calculated flood probability.

Relationships Between Entities

The entities are connected through relationships that maintain data consistency and ensure efficient database management.

User → Weather Data (One-to-Many)

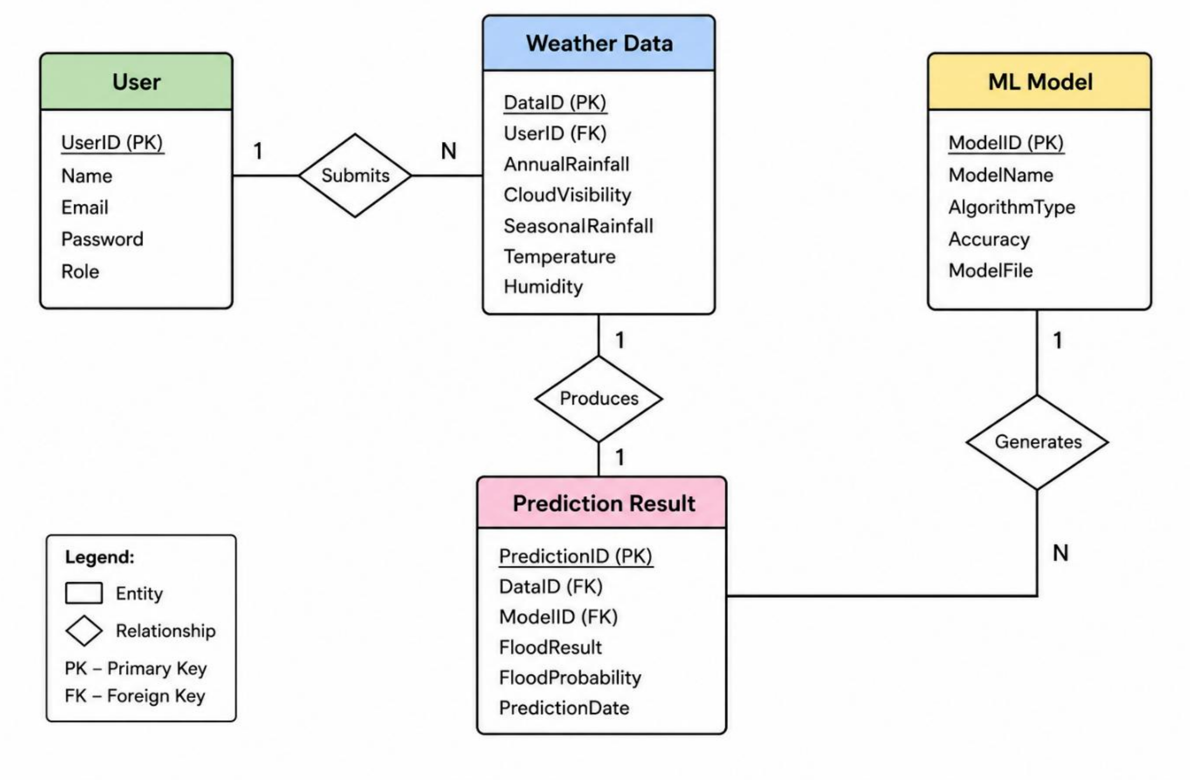
One **User** can submit multiple **Weather Data** records over time. However, each weather data record belongs to only one user. Therefore, the relationship between the **User** and **Weather Data** entities is **One-to-Many (1:N)**.

Weather Data → Prediction (One-to-One)

Each **Weather Data** record is processed by the machine learning model to generate exactly one **Prediction**. Likewise, every prediction corresponds to only one weather data record. Therefore, the relationship between **Weather Data** and **Prediction** is **One-to-One (1:1)**.

Machine Learning Model → Prediction (One-to-Many)

A single **Machine Learning Model** can process multiple weather datasets and generate many prediction records. Each prediction, however, is generated by only one machine learning model. Hence, the relationship between the **Machine Learning Model** and **Prediction** entities is **One-to-Many (1:N)**.



Advantages of the ER Diagram

The ER diagram provides several benefits during database development and system implementation.

- Provides a clear blueprint for database design.
- Reduces data redundancy through proper normalization.
- Ensures consistency and integrity of stored data.
- Simplifies database implementation and maintenance.
- Improves query performance by organizing data efficiently.
- Supports future scalability by allowing additional entities to be added without major structural changes.
- Enhances communication among developers, database designers, and project stakeholders.
- Supports efficient storage and retrieval of machine learning input and prediction data.

Role of the ER Diagram in the RISING WATER System

In the RISING WATER crop recommendation system, the ER diagram acts as the backbone of the database architecture. It enables systematic storage of user information, soil characteristics, weather conditions, crop details, and prediction results. The machine learning model retrieves the required input data from the database, processes it to generate crop recommendations, and stores the prediction results for future reference. This organized database structure improves system performance, ensures reliable data management, and supports accurate crop recommendations.

The ER diagram also provides flexibility for future enhancements. Additional modules such as fertilizer recommendation, irrigation scheduling, pest detection, crop disease prediction, and yield estimation can be integrated into the existing database with minimal modifications. This makes the system scalable and adaptable to future agricultural requirements.

Conclusion

The Entity Relationship (ER) Diagram is an essential component of the RISING WATER crop recommendation system. It provides a comprehensive representation of the database structure by defining entities, attributes, primary keys, foreign keys, and their relationships. A well-designed ER diagram ensures efficient data management, eliminates redundancy, maintains consistency, and supports the smooth functioning of the machine learning-based recommendation system. It serves as the foundation for developing a secure, scalable, and high-performance agricultural application capable of delivering accurate crop recommendations to users.